R

Reactors; Chemical Reactors

There are two main distinctions between reactors, batch and continuous. In a batch reactor a certain amount of the reactants is handled at one time. In continuous reactors, the process continues indefinitely. This is the most common type of reactor in petrochemical and refinery service.

A *batch reactor* is a closed system. An example is a batch of paper pulp being made for a specific or customized application.

A *semibatch reactor* is not a closed system. This type is useful in cases such as the manufacture of certain chemicals where a volatile chemical must be added slowly to a nonvolatile chemical (examples include the manufacture of certain glycols).

Tubular reactors (either long bent tube or shell and tube) may be either batch or continuous reactors.

Continuous reactors are "at work" all the time. This means newly introduced reactants mix to some extent with products. This extent is termed *backmixing*. A tower has many plates or baffles in it and experiences less backmixing as, for instance, a tank with no plates. Continuous reactors can then be found within towers and columns. Towers may be packed or plate (bubble cap or sieve tray) type. Optimum reactor design attempts to curtail the amount of "dead space" or areas where no reaction is taking place. It is also possible to have reactants take a shorter path than is necessary for optimum reaction. This is called *shortcircuiting*.

Catalytic reactors are continuous reactors more often than not. The main subdivision types include: fluidized or fixed bed. Fixed bed types may be either tubular, bed, or multitray types. Fluidized bed types further break down into stationary or moving (recirculating) bed types and tubular (transfer tube) types. The catalyst is generally in powdered suspension and may be removed either in batches or continuously withdrawn and regenerated. In transfer tube types, the catalyst stays in suspension with the fluid flow through the tubes.

Reactor performance is measured by its divergence from ideal conditions. Plug flow means all the fluid in the reactor has the same residence time in the reactor (no mixing with fluid streams that entered the reactor at different times). Very long tube reactors with turbulent flow can approximate this condition. Perfect mixing condition means the entering fluid in the reactor is homogeneous with the material already in the vessel on a molecular scale (perfect mixing case). In segregated mixing, the mixing is not uniform and pockets of fluid behave as "minireactors."

Refineries, Petroleum*

Crude oil is the principal raw material for a petroleum refinery. It may be of natural origin (from underground geological formations) or synthetic (recovered from tar sands). Crude oil is a mixture of many hydrocarbons and, depending on its source, varies considerably in composition and physical properties. Its elementary

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* Source: Environment Canada, extracts from EPS/1/PN/4, October 1995.

composition (by mass) usually falls within the following ranges: 84 to 87 percent carbon, 11 to 14 percent hydrogen, 0 to 3 percent sulfur, 0 to 2 percent oxygen, 0 to 0.1 percent nitrogen, 0 to 1 percent water, and 0 to 0.1 percent mineral salts. Crude oil may also contain trace amounts of heavy metals such as iron, arsenic, chromium, vanadium, and nickel.

Crude oils are broadly classified by hydrocarbon composition as paraffinic, naphthenic, asphaltic, mixed (contains paraffinic and asphaltic material), and aromatic base (prevalent in the Middle East).

The major steps in converting crude oil to various products are *separation*, *conversion*, *treatment*, and *blending*. In the first step, crude oil is separated into selected fractions mainly by distillation and to a lesser extent by solvent extraction and crystallization. Conversion processes are then used to change the size and shape of the hydrocarbon molecules to increase their monetary value. These processes include breaking molecules into smaller ones (catalytic cracking), rearranging molecules (catalytic reforming and isomerization), and joining molecules together (alkylation and polymerization). Impurities such as sulfur, nitrogen, and oxygen compounds that end up in intermediate products are removed or modified by treatment processes such as desulfurization, denitrification, or treatment with chemicals (caustic soda or acid). In the final step, the refined products are usually blended and some additives are added to improve the quality to meet finished product specifications.

These processes are discussed in more detail in the following subsections. A simplified flow diagram of the various refinery processes and products is provided in Fig. R-1.

Refinery Processes

Separation

Atmospheric distillation. In this process, the crude oil is preheated and mixed with water in a desalter. The water is then separated from the crude, taking with it the salts entrained in the oil from the geological formation. The desalted crude oil is heated and fed to the distillation column at slightly above atmospheric pressure. Next, the crude oil is separated, by distillation and steam stripping, into fractions in a range of specific boiling temperatures. The various fractions are continuously drawn off and diverted for further processing or used as finished products. The lighter products are withdrawn from the top of the column whereas lower points on the tower draw off progressively heavier fractions. The tower bottoms, which contain the heaviest petroleum fraction, are transferred to a vacuum distillation tower for further separation.

Vacuum distillation. In this process, the residue from the atmospheric distillation tower is separated under vacuum into one or more heavy gas oil streams and heavy residual pitch.

Conversion

Cracking processes. Typical cracking processes include catalytic cracking, hydrocracking, and visbreaking or coking, both of which are thermal cracking processes.

1. *Catalytic cracking* is a key process used to increase the quality and quantity of gasoline fractions. The most commonly used process is the fluid bed type, which uses a finely powdered zeolite catalyst that is kept in suspension in the reactor

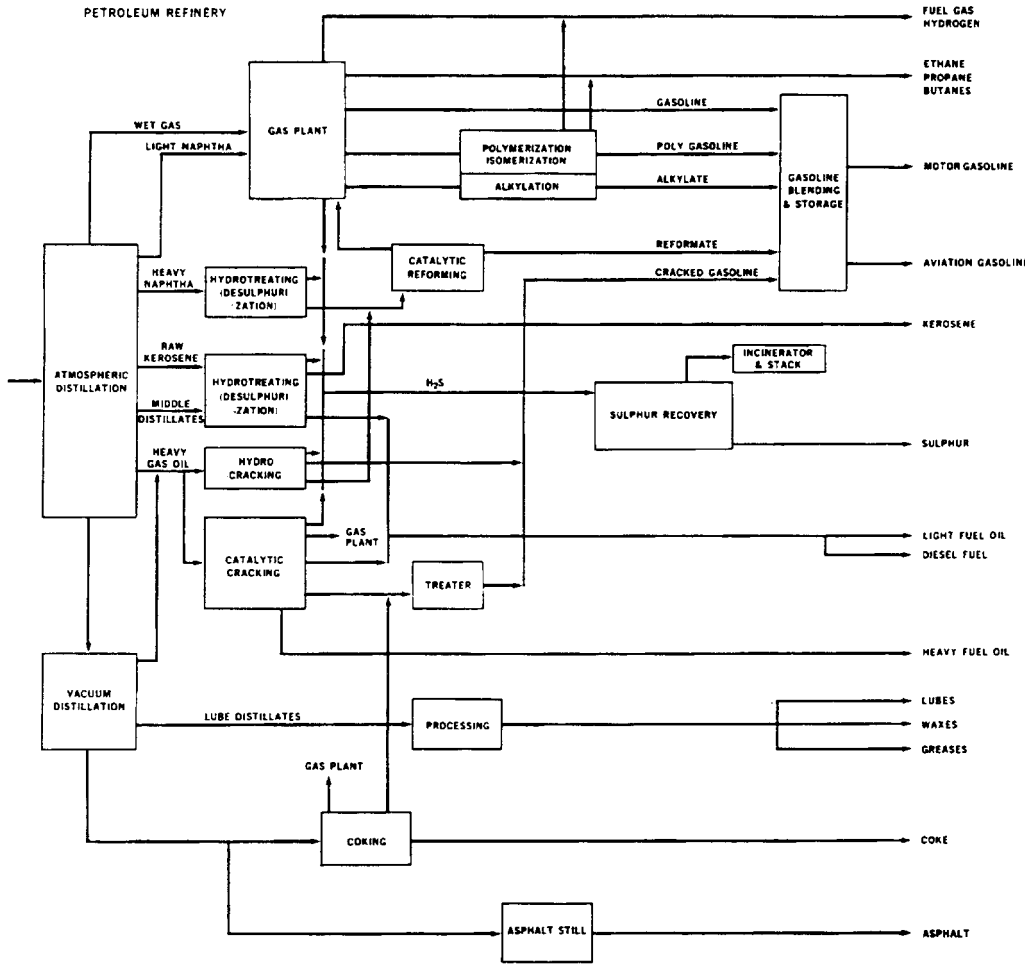


FIG. R-1 Simplified petroleum refinery process flow diagram. (Source: Environment Canada.)

by the incoming oil feed from the bottom of the reactor. Upon contact with the hot catalyst, the oil vaporizes and is cracked into smaller molecules. Vapors from the reactor are separated from the entrained catalyst and fed into a fractionator, where the desired products are removed and heavier fractions are returned to the reactor. The catalyst is deactivated by thermal degradation and through contact with heavy metals in the feed, necessitating regeneration or replacement.

2. *Hydrocracking* is basically a catalytic cracking and a hydrogeneration process. In this process, polycyclic compounds are broken to produce single ring and paraffin-type hydrocarbons. In addition, sulfur and nitrogen are removed to produce hydrogen sulfide and ammonia. These reactions occur at high temperatures and pressures, in the presence of hydrogen and a catalyst.
3. *Visbreaking* is an old process that was replaced by catalytic cracking and hydrocracking. It involves a mild thermal cracking operation designed to reduce the viscosity of the charge stock. The feed is heated and thermally cracked in the furnace. Cracked products are routed to a fractionator where the low boiling materials are separated into light distillate products, while the heavy portion may be used for coker feed or as plant fuel.

4. *Coking processes* (fluid or delayed) are used by only a few refineries in Canada. Coking is a severe thermal cracking process in which the feed is held at high cracking temperature and low pressure so that coke will form and settle out. The cracked products are sent to a fractionator where gas, gasoline, and gas oil are separated and drawn off, and the heavier material is returned to the coker.

Rearranging processes. Catalytic reforming, which is the most widely used rearranging process, improves the octane quality of gasoline obtained from crude oil. This is achieved by molecular rearrangement of naphthenes through dehydrogenation and of paraffins through isomerization and dehydrocyclization. The reformer catalyst, commonly platinum chloride on an alumina base, may also contain an activity-increasing noble metal such as rhenium. In many units, the catalyst is regenerated or replaced every 6 to 12 months. In other units, the catalyst is withdrawn continuously and regenerated on-site for further use. Refineries are more often choosing continuous reformers that do not require periodic shutdown for catalyst regeneration as conventional reformers do. The dehydrogenation and dehydrocyclization reactions produce large amounts of hydrogen as a by-product that can be used for various hydrogen-treating processes.

Combining processes. Two processes, alkylation and polymerization, are used to produce gasoline-blending stocks from the gaseous hydrocarbons formed during cracking processes.

1. *Alkylation* is the reaction of an olefin with an isoparaffin (usually isobutane) in the presence of a catalyst (either 98 percent sulfuric acid or 75 to 90 percent hydrofluoric acid) under controlled temperatures and pressures to produce high octane compounds known as alkylate. These products are separated in a settler where the acid is returned to the reactor and the alkylate is further processed. This hydrocarbon stream is scrubbed with caustic soda to remove acid and organically combined sulfur before passing to the fractionation section. Isobutane is recirculated to the reactor feed, the alkylate is drawn off from the bottom of the debutanizer, and the normal butane and propane are removed from the process.
2. *Polymerization* is a reaction that joins two or more olefin molecules. The use of this process has been declining as both the yield and quality of the gasoline product are inferior to those derived from the alkylation process. The feed must first be treated with caustic soda to remove sulfur compounds and then with water to remove nitrogen compounds and excess caustic soda. These treatments are required to protect the catalyst in the reactor. After treatment, the hydrocarbon feed is contacted with an acid catalyst in the reactor under high temperature and pressure. The catalyst is usually phosphoric acid or, in some older units, sulfuric acid. The polymerized product from the reactor is then treated to remove traces of acid.

Treating

Hydrotreating. Hydrotreating is a relatively mild hydrogenation process that saturates olefins and/or reduces sulfur, nitrogen, and oxygen compounds, along with halides and trace metals present in the feed, without changing the boiling range of the feed. This process stabilizes the product by converting olefins and gum-forming unstable diolefins to paraffins and also improves the odor and color of the products. Although there are various types of hydrotreating units, each has essentially the same process flow. The feed is combined with recycled hydrogen, heated to the

reaction temperature, and charged to the reactor. In the presence of a catalyst (metal-sulfide), the hydrogen reacts with the hydrocarbons to form hydrogen sulfide, ammonia, saturated hydrocarbons, and free metals. The metals remain on the catalyst and other products leave the reactor with the oil-hydrogen stream. The reactor products are cooled and hydrogen sulfide is removed, while hydrogen is returned to the system. The hydrocarbons are sent to a fractionator where the various products are separated. This process is ideally suited for the production of low sulfur diesel and furnace fuel oil.

Chemical treating. A number of chemical methods are used throughout the refinery to treat hydrocarbon streams. These can be classified into three groups: acid treatment, sweetening processes, and solvent extraction.

1. *Acid treatment* consists of contacting the hydrocarbons with concentrated sulfuric acid to remove sulfur and nitrogen compounds, to precipitate asphaltic or gumlike materials, and to improve color and odor.
2. *Sweetening processes* oxidize mercaptans to less odoriferous disulfides without actually removing sulfur. The most common sweetening processes are the Merox processes; others include the lead sulfide, the hydrochloride, and the copper chloride processes. In the Merox process, a catalyst composed of iron group metal chelates is used in an alkaline environment to promote the oxidation of mercaptans to disulfides using air as a source of oxygen.
3. *Solvent extraction* involves the use of a solvent that has an affinity for the undesirable compounds and is easily separated from the product. Mercaptans are extracted using a strong caustic solution. The solvent is usually regenerated by heat, steam stripping, or air blowing.

Gas treating. This process is used to remove the sulfur compounds from the various gaseous streams. Hydrogen sulfide (H_2S) can be extracted by an amine solution to produce a concentrated stream of H_2S that can be sent to a sulfur recovery plant.

Treatment by physical means. Physical methods are intermediate steps in crude oil processing operations and are often used to treat hydrocarbon streams or remove undesirable components. These methods include electrical coalescence, filtration, adsorption, and air blowing. Physical methods are applied in desalting crude oil, removing wax, decolorizing lube oils, brightening diesel oil (to remove turbidity caused by moisture), and other processes.

Deposits and compliance assessment

Refineries are held to making reports on deposits and compliance assessment. Table R-1 is a sample of this report for the Ontario region, Canada.

Blending and additives

A number of intermediate streams, called base stocks, are blended to produce a product that will meet various specifications, e.g., specific volatility, viscosity, and octane. The blending operation involves the accurate proportioning of the base stocks along with proper mixing to produce a homogeneous product.

A number of additives are used to improve the properties of the products. For example, MMT is usually added to gasoline to increase the octane number since recent regulations forbid the use of lead in gasoline. Other additives, such as anti-oxidants, anti-icing agents, and metal deactivators, are also used.

TABLE R-1 Deposits and Compliance Assessment—Ontario Region

		Refinery		
		Esso Sarnia	Petro-Canada Mississauga	Shell Corunna
A. DEPOSITS	(All guidelines and regulated deposits are for monthly averages.)			
Yearly average of daily deposits (kg/1000 m^3 of crude oil)	Guideline Deposits Esso Sarnia, P.C. Miss., Shell	Actual Deposits	Actual Deposits	Actual Deposits
Oil and Grease	17.1	0.22	4.1	0.50
Phenols	1.7	0.02	0.01	0.01
Sulfide	0.6	0.02	0.05	0.00
Ammonia nitrogen	14.3	0.70	0.86	0.03
Total suspended matter	41.1	11.3	20.5	9.7

B. COMPLIANCE ASSESSMENT

a) Number of deposits in excess of limits set in Guidelines/Regulations	M	O	D	M	O	D	M	O	D
Oil and grease	0	0	0	0	0	1	0	0	0
Phenols	0	0	0	0	0	0	0	0	0
Sulfide	0	0	0	0	0	0	0	0	0
Ammonia nitrogen	0	0	0	0	0	0	0	0	0
Total suspended matter	0	0	0	0	0	3	0	0	1
pH			0			0			0
Toxicity			0			0			0
Total	0	0	0	0	0	4	0	0	1
Percentage by region	0	0	0	0	0	80.0	0	0	20.0
Percentage of time in compliance	100	100	100	100	100	99.7	100	100	99.9

b) Number of monthly amounts exceeding the limits by:			
0 to 24%	0	0	0
25 to 49%	0	0	0
50 to 99%	0	0	0
100 to 199%	0	0	0
Over 200%	0	0	0

M, Monthly Amount; O, One-day Amount; D, Maximum Daily Amount

Actual crude rate (1000 m^3/day)	16.3	6.1	10.4
Reference crude rate (1000 m^3/day)	19.1	5.7	11.3
Status	Existing	Existing	Existing
Number of months in operation	12	12	12
Number of tests reported	1,163	1,157	1,163

Glossary: Common Terms in the Refining Industry

Activated carbon	Carbon that is specially treated to produce a very large surface area and is used to adsorb undesirable substances.
Actual deposits	The amount of contaminants discharged in refinery effluents.
Adsorption	Attraction exerted by the surface of a solid for a liquid, or a gas, when they are in contact.

Petro-Canada Oakville			Suncor Sarnia			Esso Nanticoke			Novacor Corunna					
Guideline Deposits	Actual Deposits		Guideline Deposits	Actual Deposits		Regulated Deposits Esso, Novacor	Actual Deposits		Actual Deposits			Average Authorized Deposits		Average Actual Deposits
15.1	0.80		15.3	1.2		8.6	0.83		0.08			14.3		0.90
1.6	0.01		1.5	0.01		0.9	0.00		0.00			1.4		0.01
0.6	0.06		0.5	0.02		0.3	0.02		0.09			0.5		0.03
13.3	1.8		12.7	0.94		10.3	0.06		0.84			12.9		0.66
36.1	7.6		36.6	7.2		20.5	1.7		1.3			34.3		8.1
M	O	D	M	O	D	M	O	D	M	O	D	M	O	D
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	4
		0			0			0			0			0
		0			0			0			0			0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	5
0	0	0	0	0	0	0	0	0	0	0	0			
100	100	100	100	100	100	100	100	100	100	100	100	100	100	99.9
0			0			0			0			0		
0			0			0			0			0		
0			0			0			0			0		
0			0			0			0			0		
0			0			0			0			0		
7.2			11.5			13.3	7.1		71.9					
8.8			9.0			12.5	7.9		74.3					
Existing 6.7 + Expanded 2.1			Existing 6.9 + Expanded 2.1			New	New		Existing 49.7 + Expanded 4.2 + New 20.4					
12			12			12	12							
1,163			1,163			1,160	1,145		8,114					

Aerobic bacteria	Bacteria that require free oxygen to metabolize nutrients.
Air blowing	The process used to produce asphalt by reacting residual oil with air at moderately elevated temperatures.
Altered refinery	An existing refinery at which the primary crude oil atmospheric distillation tower was replaced after October 31, 1973.

Anti-icing additive	A fuel additive used to minimize ice formation.
Anti-knock compound	Chemical compounds added to motor and aviation gasolines to improve their performance and to reduce knock in spark-ignition engines.
Antioxidants	Chemicals added to products such as gasoline and lubricating oil to inhibit oxidation.
APHA	American Public Health Association
API	American Petroleum Institute
Authorized deposits	The amount of contaminant to be discharged with the effluent of a refinery as authorized by the federal Regulations and Guidelines.
Blowdown	Removal of liquid from a refinery vessel (storage or process) through the use of pressure. The term "blowdown" is also used to refer to the actual liquid removed.
BOD	Biochemical oxygen demand. The amount of oxygen required by aerobic microorganisms to biodegrade organic matters contained in wastewater. The BOD test is used to measure the organic content of wastewater and surface water.
BPT	Best practical treatment.
Catalyst	A substance that promotes a chemical reaction without itself being altered.
COD	Chemical oxygen demand. The amount of oxygen equivalent of the organic matter required to complete chemical oxidation in an acidic medium. The COD test is used to measure the organic content of wastewater and natural water.
Cooling tower	A large structure, usually wooden, in which atmospheric air is circulated to cool water by evaporation.
CPPI	Canadian Petroleum Products Institute
Existing refinery	A refinery that began operation prior to November 1, 1973.
Expanded refinery	An existing refinery that has declared a revised Reference Crude Rate of more than 115 percent of the initial Reference Crude Rate.
Fractionator	A cylindrical refining vessel where liquid feedstocks are separated into various components or fractions.
GVRD	Greater Vancouver Regional District.
Landfill	A location where solid waste is buried in layers of earth in the ground for disposal.
Leachate	A solution resulting from the dissolving of soluble material from soil or solid waste by the action of percolating water or rainfall.
Liquid-liquid extraction	The process whereby two immiscible liquids come in contact to allow for the soluble material in the carrier liquid to be extracted in the solvent.
LPG	Liquefied petroleum gas.
Maximum daily amount	A limit set in the federal Regulations and Guidelines for a number of parameters pertaining to refinery effluents. The refinery effluent should not exceed this limit on any day of the month.
Mercaptans	A group of organosulfur compounds having the

	general formula R-SH where "R" is a hydrocarbon radial such as CH_3 and C_2H_5. Mercaptans have strong, repulsive, garliclike odors and are found in crude oil.
Monthly amount	A limit set in the federal Regulations and Guidelines for a number of parameters pertaining to refinery effluents. This limit represents the amount that should not be exceeded in the refinery effluent on a daily average basis over each month.
New refinery	A refinery that has not commenced the processing of crude oil prior to November 1, 1973.
96-hour flow-through bioassay	A test procedure required by the federal Guidelines to evaluate the acute lethal toxicity of refinery effluent to fish. The procedure consists of exposing fish to a continually renewed effluent under controlled conditions over a 96-hour period. The percent mortality of fish is observed after the four-day period.
96-hour static bioassay	A test procedure similar to the 96-hour flow-through method but in which the effluent is not renewed during the period of test.
Octane	A number indicating the relative antiknock value of a gasoline. The higher the octane number, the greater the antiknock quality.
Once-through cooling water	Water that has been circulated once through heat exchangers in order to remove heat from process streams without coming into contact with the stream.
One-day amount	A limit set in the federal Regulations and Guidelines for a number of parameters pertaining to refinery effluents. Each refinery is allowed to exceed this limit only once during a month.
Ozonation	Water treatment method that uses ozone as an oxidant to remove pollutants, i.e., chemical pollutants present in small concentrations that are difficult to remove, or to disinfect water.
Photosynthetic action	A process by which organic compounds (mainly carbohydrates) are synthesized by chlorophyll-containing plant cells. The reaction takes place in the presence of light, carbon dioxide, and water.
Priority pollutants	A list of 129 toxic pollutants having known or suspected adverse effects on human health or the environment. The United States Environmental Protection Agency (USEPA) established this list and has the mandate, under the *Clean Water Act*, to control these pollutants in wastewater discharged to the environment.
Reference Crude Rate (RCR)	The quantity of crude oil, expressed in 1000 m^3/d, declared by a refinery and used to calculate the authorized deposits.
Residual pitch	A black, heavy residue produced in the processing of crude oil.
Sour water	Water containing impurities, mainly sulfide and/or ammonia, that make it extremely harmful.

Stripping	A process for removing the more volatile components from a mixture. Generally, the hot liquid from a flash drum or tower is passed into a stripping vessel, through which open steam or inert gas is passed to removed the liquid's more volatile components.
24-hour static bioassay	A test procedure similar to the 96-hour static method but in which the percent mortality of fish is observed after a 24-hour period.
Zeolite catalyst	A catalyst that contains any of the various silicates, e.g., hydrated aluminum and calcium (or sodium) silicates, used in catalytic cracking units.

Refrigerant(s)

Refrigeration comes under heavy government scrutiny because of the potential use of halofluorocarbons that are greenhouse gases, but many times more capable of causing greenhouse effects than CO_2 (carbon dioxide). Information on refrigerants can also be found in other sections (such as Ozone and Pollutants, Chemical) in this book.

Refrigerant Technology*

Alternative refrigerants

Although the selection criteria described above appear to be very restrictive, there are several refrigerants available that will meet or exceed all the requirements.

In the decade following the signing of the Montreal Protocol, an extensive search was undertaken for alternatives that would replace chlorofluorocarbons (CFCs). Among several groups of alternatives, two—the hydrochlorofluorocarbons (HCFCs) and the hydrofluorocarbons (HFCs)—are the most useful. The HCFCs were developed to serve as interim replacements for CFCs. They are used in existing equipment for the remainder of the equipment life and in new systems, until a permanent replacement becomes available. The HCFCs contain chlorine and therefore are still ozone-depleting substances. But their ozone-depletion potential (ODP) is less than that of CFCs. For example, the ODP of R-12 is 1.0, while one of its interim replacements, R-409A, a blend of three HCFCs (R-22, R-124, and R-142b), has an ODP of 0.05. Some of the HCFCs, such as R-22, offer excellent performance characteristics and were in use even before the ozone depletion issue was raised. HFCs were developed to offer long-term alternatives to CFCs and HCFCs. They do not have any chlorine and hence have zero ODP. Many long-term alternative refrigerants are mixtures of two or more HFCs.

Refrigerant mixtures

Widely used in the past, refrigerant mixtures have received renewed interest from designers in the search for new alternatives. Mixtures offer the most attractive solution, since by mixing two or more refrigerants a new working fluid with the desired characteristics can be created. For example, by adjusting the composition of a blend containing a high-pressure and a low-pressure refrigerant, the vapor pressure of the final fluid can be tailored to match that of the CFC or HCFC being replaced. By blending refrigerants, it is possible to create new blends that are nonflammable but still contain moderately flammable refrigerants. In other cases,

* Source: Adapted from extracts from Gopalnarayanan, "Choosing the Right Refrigerant," *Mechanical Engineering*, ASME, October 1998.

blends are created to improve system characteristics, such as compressor discharge temperature, or to improve lubricant circulation by adding a more lubricant-miscible refrigerant to the blend.

Refrigerant mixtures fall into two major groups, zeotropes and azeotropes. In zeotropic mixtures, as the name implies, the liquid-vapor phase change does not occur at a constant temperature (at a fixed pressure) as in the case of pure fluids, but over a range of temperatures. Azeotropic mixtures, on the other hand, boil at a single temperature, much as a pure fluid does. The temperature alteration during phase change is commonly called temperature glide. An intermediate class of mixtures, dubbed near-azeotropes, are really zeotropes that are close to an azeotropic composition, and have a very small temperature glide. In some cases, a pair of fluids may be called near-azeotropes if the maximum temperature glide for the pair is very small, even though they may not form an azeotrope. A 50/50 mixture of R-32 and R-134a, for example, forms a zeotropic blend with a temperature glide of about 5°C at about 950 kPa. A 50/50 blend of R-115 and R-22 will form a perfect azeotrope, while a 50/50 blend of R-32 and R-125 will form a near-azeotrope with a temperature glide of about 0.1°C.

The higher temperature glides in zeotropes, in certain applications, offer the potential for higher efficiencies. By more closely matching the temperature profiles of the refrigerant blend and the fluid being cooled, engineers can reduce the external heat transfer irreversibilities. But this advantage cannot be fully utilized, because there is now an increased mass-transfer resistance in the process of boiling and condensation. The temperature glide also results in a phenomenon called fractionation, whereby higher and lower boiling components tend to separate; the vapor will be richer in the fluid with a lower boiling point and vice versa. In the event of a vapor leak in the system, the leaking vapor will be richer in the more volatile component of the blend. This could result in a change in the mixture's composition, which is undesirable, since it could alter the system's performance. However, performance changes will result even for pure fluid if there is a refrigerant leak, since change in refrigerant inventory will cause a change in performance. Class 1 zeotropic blends are formulated to be nonflammable even under the worst fractionation scenario.

Azeotropes and near-azeotropes, on the other hand, do not have a temperature-glide-induced problem. However, azeotropes and near-azeotropes that match the vapor-pressure characteristics of the CFC or HCFCs they replace are nonexistent. Hence, as can be seen in Fig. R-2, a multitude of alternative blends are zeotropes.

Figure R-2 shows a broad outline of different applications, the refrigerants that were used in the past, and the interim and long-term alternatives available. According to the American Society of Heating, Refrigerating, and Air-Conditioning Engineers' (ASHRAE's) refrigerant-numbering scheme, refrigerants having the form R-4XXX are zeotropic blends of two or more refrigerants, while those with the form R-5XXX are azeotropes. R-134a was a natural answer as a long-term R-12 replacement, since it met almost all the selection criteria that were imposed. Although not compatible with many of the elastomers or lubricants used with R-12, it provided the single-fluid solution, and the issues of compatibility and miscibility were addressed by selecting alternative elastomers and lubricants, respectively. The changeover to R-134a in place of R-12, however, required a concerted effort and considerable expense on the part of equipment manufacturers.

Comfort conditioning

R-22 is being used extensively in comfort conditioning. (See Fig. R-2.) Unlike R-12, however, there is no known single-fluid answer that would meet all the selection criteria. One of the alternatives proposed is propane. Although propane has a low global warming potential (GWP) compared to other alternatives, it may have a

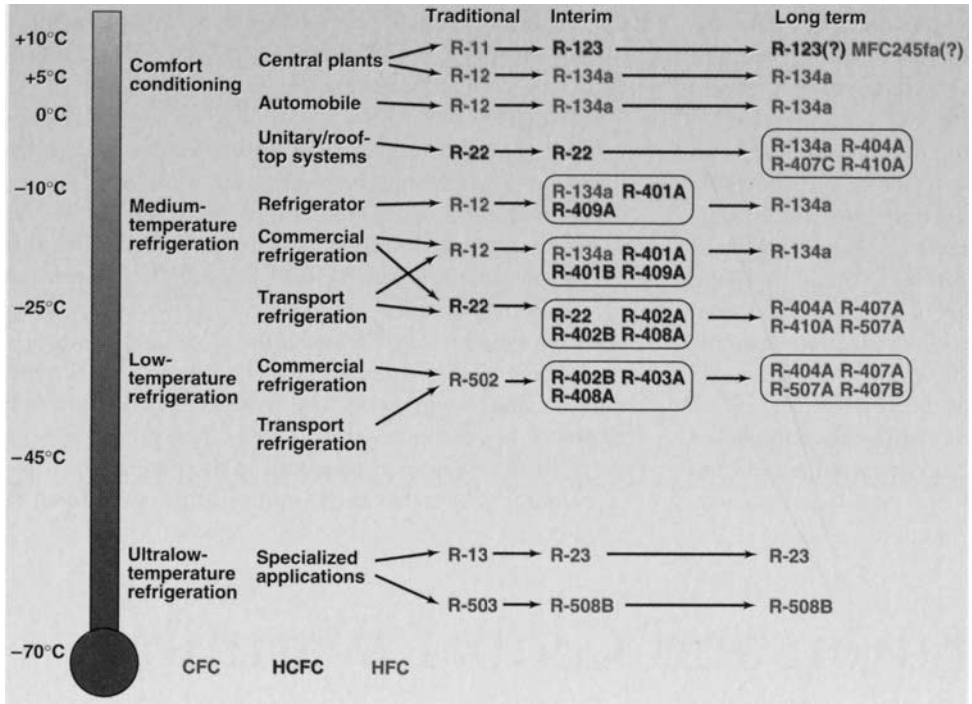


FIG. R-2 For a variety of applications, a number of interim and long-term replacements are available. (Source: Gopalnarayanan.)

higher total equivalent warming impact (TEWI). In addition, being flammable, it will require additional safeguards, resulting in a more expensive system.

Two alternatives that have become acceptable as replacements for R-22 in comfort-conditioning applications, R-407C and R-410A, are both refrigerant mixtures containing HFCs. HFCs are immiscible with conventionally used mineral or alkyl benzene lubricants, and so require either polyalkylene glycol or polyol ester lubricants.

R-407C is a ternary zeotropic mixture of R-134a, R-125, and R-32. One of the advantages of R-407C is that its vapor pressure is only slightly higher than that of R-22, so it can be used as a "retrofit" fluid in R-22 systems with only a lubricant change. It also has performance comparable to R-22. One disadvantage of R-407C is its temperature glide, which can be as high as 6 to 7°C at typical system pressures, leading to problems related to the fractionation discussed earlier.

The other alternative, R-410A, is a near-azeotropic mixture of R-32 and R-125. The vapor pressure of this fluid is almost 50 percent higher than that of R-22, and thus R-410A cannot be used in existing systems. Newly built systems will have to be redesigned to handle the increased pressures. However, the higher pressure and the fact that the fluid is almost an azeotrope provide an opportunity to design more compact systems with greater efficiency.

Which is a better fluid? Again, the choice depends on the application and local regulations. For example, in certain European countries, the phaseout date for R-22 use in new equipment was 1998 (as opposed to 2010 in the United States). In this instance, there may not be a newly designed R-410A unit available for changeover. In some cases, it may not be economically viable to redesign the entire system, considering that even the manufacturing plant will require redesign, while in others, the opportunity for better system performance with R-410A may justify the investment.

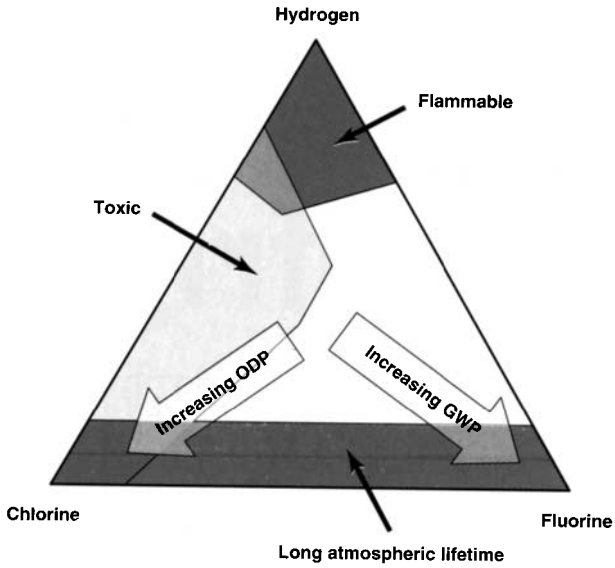


FIG. R-3 When considering appropriate halocarbon alternatives, engineers need to strike a balance between key constraints. (Source: Gopalnarayanan.)

Striking a balance

One of the major challenges in identifying halocarbon alternatives to ozone-depleting substances is to strike a balance between the various affecting factors, namely ODP, TEWI, toxicity, flammability, performance characteristics, and cost (see Fig. R-3). Due to halocarbons' level of GWP, some European countries have begun exploring alternatives to them. Certain naturally occurring compounds, such as hydrocarbons, carbon dioxide, and ammonia, would need significant processing or refining to be used as refrigerants. These processed refrigerants, as well as natural refrigerants such as air or water, continue to be considered and in some cases applied as viable alternatives to synthetic refrigerants. In the United States, there is considerable resistance to use of hydrocarbons or ammonia in applications that use halocarbons. This is due to liability concerns arising from their flammability and, in the case of ammonia, toxicity. Even considering their higher direct GWP, halocarbon refrigerants appear to offer the most complete solution, by fully meeting all the safety, performance, economic, and environmental constraints that are placed on today's refrigerants.

Refrigeration and global warming

If HFCs are the obvious solution to the refrigerant quandary, why is there so much activity in developing alternative refrigerants and technologies? One reason, it is thought, is the higher GWP of HFCs.

For purposes of comparison, R-134a has a GWP of about 1300, while that of the hydrocarbon refrigerant isobutane is about 11. Both of these refrigerants are used as replacement fluids for R-12 in domestic refrigerators. Is the difference in GWP significant? The answer lies in realizing that the contribution to global warming comes not only from the refrigerant that may leak from the system, but also from the amount of carbon dioxide that will be emitted at the power plant in producing the energy needed to run the compressor.

The effect of a refrigeration system on global warming is more accurately described by the popular concept of TEWI, which takes into account both of these

contributions to global warming. TEWI, therefore, ties in the energy efficiency of the system (and that of the power plant) in determining the effect on global warming. In the case of the refrigerator using R-12, only 4 percent of the TEWI comes from direct emission of the refrigerant. Considering that R-134a has only about one-fifth of the GWP of R-12, the contribution to TEWI of using either R-134a or isobutane makes a difference of only 1 percent over the lifetime of the refrigerator.

It is estimated that HFCs currently account for only 0.06 percent of human-induced global warming. Even assuming that HFCs will be fully emitted from the systems in which they are used, estimates place their contribution at about 0.6 percent in 2010, less than 2 percent in 2030, and approximately 2.6 percent by 2100.

Approaches to the global-warming issue taken by various countries are different. In the United States, the consensus is to reduce TEWI by attacking the energy efficiency of the equipment, since that seems to contribute the most to the TEWI. In some European countries, however, emphasis is placed only on reducing the direct contribution from the refrigerants. As in the case of politically sensitive issues like this, the approaches taken by various governments are not necessarily always based only on technical merit.

Natural Refrigerants*

The search for new and environmentally benign refrigerants to replace the existing CFCs and HCFCs has led to the introduction of HFCs. However, HFCs have a much higher global-warming potential and higher costs than natural refrigerants. These concerns have spurred calls for the investigation of alternatives to HFCs. Some environmentalists would like the refrigeration industry to bypass HFCs and employ natural refrigerants as soon as possible.

Natural refrigerants are working fluids based on molecules that occur in nature. Examples are such substances as air, water, ammonia, hydrocarbons, and carbon dioxide. Nevertheless, the actual fluids used in refrigeration systems may very well be synthesized and will not necessarily be extracted from nature. Ammonia, for instance, is synthesized in large quantities, and hydrocarbons undergo an extensive chemical processing procedure. Still, the cost of these fluids is much lower than that of HFC refrigerants, and they do not affect the environment in an unknown way. Also, the amount of fluid produced is negligible compared with the amount available in nature.

Studies of natural refrigerants continue. For example, Annex 22 of the International Energy Agency implemented a three-year project, Compression Systems with Natural Working Fluids, in 1995. Air, water, ammonia, hydrocarbons, and carbon dioxide have a low or zero direct global-warming potential and zero ODP, as shown in Table R-2.

Development of refrigeration systems

Since the first vapor-compression refrigeration system was invented by Jacob Perkins in 1834, more than 50 chemical substances have been used as refrigerants in refrigeration and air-conditioning systems. The development of refrigeration systems using CO_2 as the refrigerant started in 1866, when an ice production machine that used CO_2 was invented by Thaddeus S.C. Lowe. In 1880, the first CO_2

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*Source: Adapted from extracts from Hwang, Ohadi, and Radermacher, "Natural Refrigerants," *Mechanical Engineering*, ASME, October 1998.

TABLE R-2 Characteristics and Properties of Some Refrigerants

Refrigerants	R-12	R-22	R-134A	NH_3 (R-717)	C_3H_8 (R-290)	CO_2 (R-744)
Natural substance	No	No	No	Yes	Yes	Yes
TLV (ppm)	1,000	1,000	1,000	25	1,000	5,000
IDLH (ppm)	50,000	—	—	500	20,000	50,000
Amount per room volume (vol/kg/m^3)	4.0/0.2	4.2/0.15	—	—	0.44/0.008	5.5/0.1
Flammable or explosive	No	No	No	Yes	Yes	No
Flammability limits in air (volume%)	—	—	—	15.5/27	2.2/9.5	—
Toxic/irritating decomposition products	Yes	Yes	Yes	No	No	No
Approximate relative price	1	1	3–5	0/2	0.1	0.1
Molar mass	120.9	86.5	102.0	17.0	44.1	44.0
Refrigeration capacity at 0°C (kJ/m^3)	2,740	4,344	2,860	4,360	3,870	22,600

TLV: Threshold limit value (the refrigeration concentration limit in air for a normal 8-hour workday; it will not cause an adverse effect on most people).
IDLH: Immediately dangerous to life or health (maximum level from which one could escape within 30 minutes without impairing symptoms or any irreversible health effects).

compressor was designed by Franz Windhausen. After the late 1800s, the use of CO_2 refrigeration systems increased. As a result of continuous efforts to improve efficiency, two-stage CO_2 machines were developed in 1889 by the J. & E. Hall Co. in Great Britain, and the multiple-effect CO_2 cycle was developed by G. T. Voorhees in 1905.

Meanwhile, Thomas Midgley, Jr., and Albert Henne published a paper in 1930 on fluorochemical refrigerants, as a result of a search for stable, nontoxic, nonflammable, and efficient refrigerants. In 1931, dichlorodifluoromethane, CFC-12, was commercially produced. After the introduction of fluorochemical refrigerants, the early refrigerants, including CO_2, were replaced by many other CFCs and HCFCs. This led to a drastic decline in the use of refrigerants other than CFCs and HCFCs after World War II. Only ammonia remained in use, and its application was confined to large industrial systems. In recent years, though, natural refrigerants have gained considerable attention as alternative refrigerants for mobile air conditioners in Europe.

Varieties of natural refrigerants

Although natural refrigerants were used extensively in the early years of refrigeration technology, a number of technical and safety challenges caused them to be abandoned when CFCs became available. These challenges still exist today for air, water, ammonia, hydrocarbons, and carbon dioxide.

Air is used extensively as a refrigerant in aircraft cooling. Its advantages are that open systems require fewer heat exchangers, aircraft have compressed air available already, and systems tend to be low in weight. Its efficiency, however, is quite poor. Nonetheless, German railways have installed air cycles in the latest generation of high-speed trains because of weight concerns and, most importantly, because maintenance time is very short. There is no cumbersome and time-consuming refrigerant reclamation and no system evacuation is required.

Water has the potential to be a very efficient refrigerant, but because it requires operation in a deep vacuum, its vapor density is quite low. This leads to costly large-volume vacuum tanks that must house all the machinery, such as heat exchangers and compressors. Furthermore, water's pressure ratio is very high, imposing additional challenges for compressors that must operate in a deep vacuum.

The only applications where water is used as a refrigerant on a commercial basis are in large-capacity lithium-bromide water-absorption chillers. Over the last

decade, a few large-scale water-vapor compression systems have been used commercially. One such system provides cooling to the LEGO factory in Denmark, another to a mine in South Africa. In both cases, open systems are employed, and the need for heat exchangers is eliminated by using direct-contact heat exchange. Thus, the chilled water that circulates through the facility is also used as the refrigerant. Although various research projects are underway worldwide, a demanding compressor technology, the need for vacuum pumps, and degassing remain great challenges in terms of cost and design.

Ammonia is also a very good refrigerant and is used to a significant extent in large warehouses. Ammonia is toxic and, under certain limited conditions, flammable and even explosive. However, with its intense, pungent odor, it is a self-alarming refrigerant. Ammonia has emerged as a refrigerant for water chillers in Europe. These units are entirely self-contained, including a gas-tight cabinet that houses the entire unit and a water tank to dissolve any ammonia in case of a leak. These measures, to be sure, increase costs considerably.

Hydrocarbons are excellent refrigerants, but they are also flammable and explosive. In North America, any flammability risk is unacceptable, but some countries in Europe and elsewhere have less-stringent liability laws. Since the mid-1990s, virtually all refrigerator production in Germany has used hydrocarbons as the working fluid. Some heat pump manufacturers whose systems are installed entirely outdoors have followed suit, and some commercial installations have recently become publicly known. Nevertheless, the danger of fire remains an overriding concern. To address this challenge with safety features, the cost of a system has to be increased by about one-third.

Carbon dioxide is a refrigerant that operates at very high pressures in a transcritical cycle for most operating conditions. Thus the refrigerant condenser of a conventional refrigeration system serves now as a cooler for supercritical fluid. Only after the expansion process is liquid carbon dioxide available to provide cooling capacity through evaporation. Because of the nature of the transcritical cycle, the efficiency of carbon dioxide is quite poor. However, this is its only disadvantage. All the other characteristics of carbon dioxide are very favorable. It is environmentally safe, has very low toxicity, and allows for extremely compact systems. The vapor pressure of CO_2 is approximately seven times higher than that of R-22. Moreover, the supercritical CO_2 has a higher density than subcritical fluids, so there is potential to reduce the size of hardware. There are indications that with modern materials and technologies, the weight of CO_2 heat exchangers can be reduced considerably, especially for tap water heating, with essentially the same performance.

Gustav Lorentzen and his colleagues revived research on the CO_2 cycle in the 1990s, especially for mobile air-conditioning applications. This group has focused on the experimental evaluation and thermodynamic modeling of mobile air-conditioning systems. Meanwhile, a group of researchers in Germany has focused on railway air systems. In 1994, European automobile manufacturers launched a joint research project, Refrigeration and Automotive Climate Systems under Environmental Aspects (RACE), to investigate an air-conditioning system with the natural refrigerant CO_2. Most efforts conducted so far with respect to CO_2 have focused on experimental evaluation and thermodynamic modeling for R-12 and R-22 replacement.

Investigations of prototypes for automotive air conditioning and some other applications show that well-designed systems can actually perform at levels reached by other refrigerants. One method that would greatly enhance the efficiency of CO_2 cycles is the use of an expander instead of an expansion valve. An expander produces work that can be fed back into the compressor or used otherwise. In

TABLE R-3 Thermophysical Properties of CO_2 at –25/–10/5°C

Refrigerant	R-22	CO_2 (R-744)
Saturated pressure (MPa)	0.201/0.354/0.584	1.683/2.649/3.969
Latent heat (kJ/m^3)	223.5/212.8/201.0	293.2/258.6/215.0
Saturated liquid density (kg/m^3)	1,362.0/1,314.7/1,264.3	1,054.2/982.9/927.4
Saturated vapor density (m^3/kg)	8.96/15.3/24.8	43.9/71.2/114.6
Saturated liquid specific heat (kJ/kg-K)	1.113/1.144/1.184	2.113/2.306/2.726
Saturated vapor specific heat (kJ/kg-K)	0.649/0.699/0.761	1.221/1.519/2.160
Saturated liquid viscosity (μPa-s)	286.1/242.6/206.3	156.8/123.8/95.4
Saturated vapor viscosity (μPa-s)	10.38/11.01/11.65	12.76/13.83/14.75
Saturated liquid thermal conductivity (mW/m-K)	113.6/105.4/97.3	141.3/123.2/105.3
Saturated vapor thermal conductivity (mW/m-K)	8.32/9.02/9.79	14.8/17.7/22.8
Surface tension (mN/m)	15.5/13.2/11.0	9.6/6.4/3.5

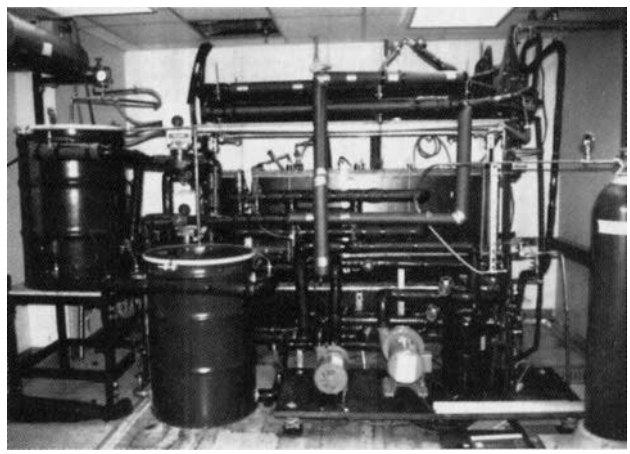


FIG. R-4 This is the experimental loop for a recent CO_2 cycle performance and heat-transfer study done at the University of Maryland in College Park. (Source: Hwang, Ohadi, and Radermacher.)

conventional systems, the expander is beginning to emerge in large-capacity units. However, any kind of machinery added to a cycle will increase cost to an unacceptable degree. Therefore, most research and development projects underway today do not include an expander.

R-22 and carbon dioxide

Thermophysical properties of R-22 and CO_2 are compared for the saturation temperatures of –25, –10, and 5°C in Table R-3. The evaporation pressure of CO_2 is four to eight times higher than that of R-22. The saturated liquid density of CO_2 is approximately 70 percent that of R-22, while the saturated vapor density of CO_2 is approximately five times that of R-22. The higher density offers the opportunity to reduce heat exchanger size and weight. CO_2 has better heat-transfer characteristics (higher latent heat, specific heat, and thermal conductivity) and lower viscosity than R-22. See Fig. R-4.

The first approach to comparing the refrigerants is to compare their ideal cycle performance. An ideal cycle is defined as a refrigeration cycle that has zero approach temperatures and minimum or no pressure drops in the heat exchangers. It should also have isentropic compression, no subcooling (for R-22), and no suction superheating.

This ideal cycle is applied to an R-22 refrigeration simulation model (REFSIM) and a CO_2 simulation model (CO_2SIM). Overall, the ideal-cycle coefficient of performance (COP) of CO_2 is only 50 to 60 percent of R-22's at various chilled water temperatures and various gas-cooler cooling water temperatures.

However, the COP of the actual cycle for the CO_2 water chiller is 94 to 106 percent of the existing R-22 water chillers, based on recent experimental work. This difference between the ideal cycle and actual cycle is analyzed using refrigeration cycle models. The actual cycle is different from the ideal cycle because of the irreversibility of the actual process, which can be attributed to the following parameters: compressor efficiency, approach temperature in heat exchangers, pressure drop in heat exchangers, and degree of subcooling and superheating. In a comparison of the effects of these parameters on the cycle COP change for both refrigerants, the compressor efficiency for both refrigerants has the largest impact on the COP drop. Moreover, this parameter has more impact for R-22 than CO_2, in absolute terms. The effects of the suction gas heating by the motor, as well as mechanical losses, are much less for CO_2 than R-22, due to the higher specific heat of CO_2. The approach temperature and degree of subcooling in the condensing process have the second and third largest impacts on the COP drop of R-22. This difference is a result of the better heat transfer of CO_2. The overall heat-transfer coefficient of CO_2 during the gas-cooling process is approximately double that of R-22 during the flow-condensation process. Another important parameter is the sensitivity of temperature change to the pressure drop. CO_2 is less sensitive than R-22. The more sensitive case has a larger approach temperature and COP drop.

Thus, it can be concluded that the beneficial thermophysical properties of CO_2 contribute considerably to its good overall COP. It is therefore conceivable that CO_2 may be a promising candidate as a viable refrigerant. Extensive research work is required to prove feasibility and, if successful, an industrywide development effort will be required to make this fluid succeed.

Natural refrigerants still have a number of technical and safety challenges to overcome, and each has its unique advantages and disadvantages. Although it is not clear yet which one is the dominant candidate as an alternative refrigerant, CO_2 offers a clear advantage over CFCs and HCFCs from the environmental impact (ODP and GWP) standpoint. In addition to its environmental advantages, CO_2 also offers certain attractive thermal characteristics that can help it provide substantial potential as a long-term replacement if energy efficiency challenges can be addressed.

Design Constraints*

There are five major factors that design engineers must consider in selecting a refrigerant for a particular application: performance, safety, reliability, environmental acceptability, and simple economics. In assessing these areas, design engineers should keep in mind that the relative importance of each depends upon the application and, of late, government regulations, which differ in each country.

Two of the primary criteria in the performance of a refrigeration system are refrigeration capacity and efficiency. The refrigeration capacity is the amount of cooling that the system can produce for a given volumetric flow rate of the refrigerant. Volumetric capacity is dependent not only on the latent heat of

* Source: Adapted from extracts from Gopalnarayanan, "Choosing the Right Refrigerant," *Mechanical Engineering*, ASME, October 1998.

vaporization but also on the density of the refrigerant vapor that enters the compressor.

One good indicator of a refrigerant's capacity is its normal boiling point. The higher the boiling point, the lower the fluid's volumetric cooling capacity. On the other hand, fluids with higher boiling points tend to have greater efficiency. However, they also tend to have a higher pressure drop and, in some cases, lower heat-transfer coefficients. Thus there is a need for compromise. Other refrigeration system requirements include appropriate operating pressure, smaller compressor size, and lower compressor-discharge temperature.

As far as safety is concerned, toxicity, flammability, and pressure govern the proper use of a refrigerant. Refrigerants are broadly classified, based on their toxicity and flammability, by ASHRAE Standard 34-1997. For toxicity, the classification is based on the chronic exposure limits (the level to which an individual can be exposed over his or her working life without ill effects), which are defined by the threshold limit value (TLV), the maximum exposure at any given time, and the time-weighted average (TWA). Refrigerants with no identified toxicity at concentrations less than 400 parts per million (ppm) based on TLV and TWA are considered Class A, while those that show evidence of toxicity at concentrations below 400 ppm are Class B. Most of the refrigerants in use are Class A. However, there are a few Class B refrigerants, such as R-123 and ammonia, that need additional equipment safety.

Regenerator

This term is normally used in conjunction with gas turbines. The thermal efficiency of a gas turbine can be increased by using the exhaust heat from the turbine to heat the gas-turbine compressor discharge air. *See also* Cogeneration; Turbines, Gas.